

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Comprehensive Test
(EC-3 Regular)

Course No. : AIMLZ565
Course Title : Machine Learning
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 24/05/2026 (FN/AN)

No. of Pages	= 6
No. of Questions	= 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Please Note:

A maximum of 12 main questions with Q.Nos. 1 to N. **Please do not include any objective type questions. Please do not include any reference material like data tables in the question paper, as all Mid-Semester Test are completely Closed Book examinations. Both the options to submit answers i.e. typing as well as scanning & uploading hand written answers should be available for ALL the questions.**

Q.1

- (a) **[6 Marks]** Given the following dataset, classify point $P = (2,2)$ using k-nearest neighbor method with $k = 3$ and **Manhattan distance**.

Point	X	Y	Class
A	1	2	Red
B	2	3	Red
C	3	3	Blue
D	6	5	Blue

- (b) **[4 Marks]** Suppose we wish to use the K-Nearest Neighbors (kNN) method for a classification problem. Explain how the output prediction is computed for a new instance. Explain the typical effect of a low value (eg. $k=1,2$ etc.) and a high value (eg. $k=10,20$ etc.) on the following

- **Sensitivity to noise** : i.e. whether low/high values of k result in a low/high sensitivity to noise
- **Bias-Variance Tradeoff** i.e. whether low/high values of k result in a low/high bias, low/high variance

Q.2 A medical clinic tests patients for a disease using a quick blood test. Assume that the following are known:

- 2% of the population actually has the disease.
- If a person HAS the disease, the test returns Positive 90% of the time. (Sensitivity)
- If a person does NOT have the disease, the test returns Positive 5% of the time. (False Positive Rate)

Suppose we define events as below:

D+ : 'a selected person has the disease',

D- : 'a selected person does not have the disease'

T+ : 'a selected person tests positive for the disease'

T- : 'a selected person tests negative for the disease'

(a) **[3 Marks]** Calculate and write down the values of the following probabilities. If the information provided is insufficient, enter your answer as "Insufficient information to calculate the specified probability"

- $P(D+)$:
- $P(T+ | D+)$:
- $P(T+ | D-)$:

(b) **[2 Marks]** Calculate the overall probability of testing positive i.e. $P(T+)$

(c) **[5 Marks]** A patient tests **positive for the disease**. Use **Bayes' theorem** to compute the probability that this patient actually has the disease.

Q3 A coin is tossed **8 times** and the following outcomes are recorded:

H, H, T, H, T, H, H, T

Let p = probability of getting a Head on any single toss. We would like to compute the Maximum Likelihood Estimate (MLE) of the parameter p based on the observed data above. Assume that the data is i.i.d. (independent, identically distributed) i.e. each coin toss is independent of the others and the probability of getting a Head is the same in any coin toss.

(a) [2 Marks] Write down the **likelihood function** $L(p)$ for the above observations.

Hint : Use the Binomial formula.

(b) [3 Marks] A common practice in Maximum Likelihood Estimation is to minimize the log-likelihood function. Compute the Maximum Likelihood Estimate of the parameter p by minimizing the log-likelihood function Clearly show your working and all relevant intermediate steps to get full credit.

(c) [10 Marks] The table below records past days and whether a cricket match was played.

Use the data in the table below to build a Naïve Bayes classifier and use it to predict whether cricket can be played or not for the following set of weather conditions

{ Outlook = Sunny, Temperature = Cool, Humidity = High, Wind = Strong }

Day	Outlook	Temperature	Humidity	Wind	Play Cricket?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rainy	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rainy	Mild	High	Strong	No

Q4

- (a) **[6 marks]** A hard-margin linear SVM is defined by the support vectors, Lagrange multipliers, and class labels as shown in the table below. Write the decision boundary equation by finding the weight vector w (which defines the hyperplane) and bias b . To calculate the bias use the condition that if $\mathbf{x}_i$ is a support vector and y_i is the corresponding class label, then $y_i(w \cdot \mathbf{x}_i + b) = 1$. Apply this condition for each support vector $\mathbf{x}_i$, and find the average of all the b s obtained by solving each such equation.

Lagrange multiplier	Support Vector	Class label
0.25	(0, -1, 1)	+1
0.25	(0, 1, -1)	-1

- (b) **[4 marks]** Consider a 2D dataset consisting of the following points categorized into two classes:

Positive Class ($y = +1$): (2, 2), (2, -2), (-2, -2), (-2, 2)

Negative Class ($y = -1$): (1, 1), (1, -1), (-1, -1), (-1, 1)

Examine the given dataset (draw a somewhat to scale graphical plot) and determine whether it is linearly separable in the original input space. Provide a brief justification based on the distribution of points.

- (c) **[5 marks]** Suppose we wish to devise an SVM using a dual Lagrangian formulation as below.

$$\text{Maximize } F(\Lambda) = \Lambda \cdot \mathbf{1} - \frac{1}{2} \Lambda \cdot D \Lambda$$

subject to

$$\Lambda \cdot \mathbf{y} = 0$$

$$\Lambda \leq C \mathbf{1}$$

$$\Lambda \geq \mathbf{0}$$

Let $C = 1$. Here Λ is a vector of Lagrange multipliers and D is a the kernel matrix where the (i,j) entry is given by $D(i,j) = y_i y_j K(x_i, x_j)$ where y_i is the class label for the i^{th} instance. Suppose a polynomial kernel of degree 2 where the kernel is specified by the formula $K(u, v) = (u \cdot v + 1)^2$. Compute the kernel matrix D for the above problem.

Q5 A retail analytics team is clustering customers based on two features:
 x_1 : Average monthly spend and x_2 : Visit frequency. The following 2-D dataset is obtained:

Point	A	B	C	D	E	F
x_1	1	2	2	6	7	8
x_2	2	1	3	5	6	5

a) **[6 marks]** The analytics team decides to partition the given 2-D customer data into two groups using a centroid-based clustering approach. At the start of the procedure, two representative points in the feature space, namely A and F i.e. points located at (1,2) and (8,5) are selected as the initial cluster centers. Using these initial centers as references, determine how the data points would be grouped after the first complete update cycle of the K-Means algorithm. Provide a step-by-step explanation of:

- (i) How proximity to the current centers influences membership?
- (ii) How do the representative locations of the two groups change once the memberships are determined?

Clearly state the resulting groups and the updated representative points obtained at the end of this cycle.

b) **[4 marks]** Assume the same data shown in table above is modeled using a Gaussian Mixture Model (GMM) with two components having, equal mixing weights, spherical covariance matrices and means initialized to the centroids obtained in part (a)

Explain one key difference between cluster assignments in K-Means and responsibilities in GMM, using one specific point from the dataset to justify your answer.

c) **[2 marks]** Briefly and clearly explain how clustering is performed using Gaussian Mixture Models. Specifically explain what mixing weights mean in this context.

d) **[3 marks]** State one data characteristic under which GMM would produce better clusters than K-Means for a dataset, and explain why. You could use the dataset in part a) as an example to illustrate your answer, if you choose.

Q6

(a) **[2 marks]** Explain one key difference between a Bagged Tree and Random Forest.

(b) **[3 marks]** Consider a dataset $D = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Suppose a bagged sample is $\{1, 1, 2, 4, 6, 6, 7, 8, 9, 9\}$

(i) What are the Out of Bag samples in this particular example?

(ii) Calculate the OOB error for the above sample.

If it is not possible to calculate either of the above, clearly state whether any information is missing.

(c) **[2 marks]** Suppose we use AdaBoost for a classification problem. Suppose we run AdaBoost for 4 rounds and the classifier weights and predictions for an instance are as shown in the table below. Compute the predicted class label. Clearly show intermediate steps and indicate if any information is missing.

Model	Prediction	α
1	+1	0.3
2	-1	0.7
3	+1	0.15
4	+1	0.3

(d) **[8 marks]**

i	1	2	3	4	5	6	7	8	9	10
Class label y_i	-1	-1	-1	-1	-1	1	1	1	1	1
Prediction	1	-1	-1	-1	1	1	-1	1	1	1

Consider a dataset with 10 instances as shown in the table above. Suppose we use AdaBoost for classification and the predictions in the first round are as shown in the table above. Assuming equal weights for all instances initially, **calculate the weights of the instances after round 1 (whose predictions are shown above)**. Assume that the weights of correctly classified examples are decreased exponentially and those of incorrectly classified examples are increased exponentially. Normalize the weights such that they sum to 1.